

EXPERIMENT 14:

Perform transformation using Homography matrix.

AIM:- Transformation using Homography matrix

PROGRAM:-

```
import cv2
import numpy as np
import os

image_path = r'C:\Users\pavan\OneDrive\OpenCV\tree.jpg'
if not os.path.exists(image_path):
    print("Image file not found at:", image_path)
    exit()

img = cv2.imread(image_path)
if img is None:
    print("Failed to load image. Check file format or path.")
    exit()

rows, cols = img.shape[:2]
src_points = np.float32([
    [0, 0],
    [cols - 1, 0],
    [0, rows - 1],
    [cols - 1, rows - 1]
])

dst_points = np.float32([
    [0, 0],
    [cols - 1, 0],
    [0, int(0.7 * rows)],
    [cols - 1, int(0.7 * rows)]
])

M, mask = cv2.findHomography(src_points, dst_points)
if M is not None:
    homography_img = cv2.warpPerspective(img, M, (cols, rows))
    cv2.imwrite('homography_transformed.jpg', homography_img)
    print("Transformation complete. Output saved as homography_transformed.jpg")
```

else:

```
print("Homography could not be computed.")
```

INPUT:-



Output:-



RESULT:-

Thus, the program is executed successfully